

Care & Share - Full Consolidated Report

1. Introduction

Care & Share is a web-based community platform where users can donate items, request donations, exchange products, and resell approved products. The system supports role-based access for users, NGOs, and admins. NGO accounts require document verification and admin approval before login access is granted.

2. Module Description

- Authentication Module: Register/login/logout, password reset, JWT-based access control.
- User Module: Profile and dashboard data, user activity tracking.
- Donation Module: Add donation items, browse available donations, submit donation requests.
- Product Module: Add exchange/resell/donation listings, browse approved products.
- Exchange Request Module: Submit and manage exchange requests between users.
- Purchase Module: Create and track purchase requests for resell items.
- Admin Module: Approve/reject products, donation items, donation requests, exchange requests, NGO accounts, and inventory management.
- NGO Verification Module: Dedicated NGO registration with mandatory document upload and approval workflow.

3. Hardware and Software Need

Minimum Hardware

- CPU: Dual-core 2.0 GHz
- RAM: 4 GB
- Storage: 10 GB free
- Network: Stable internet connection

Recommended Hardware

- CPU: Quad-core 2.5+ GHz
- RAM: 8 GB or more
- Storage: SSD with 20+ GB free

Software Requirements

- Operating System: Windows / Linux / macOS
- Runtime: Node.js (LTS)
- Database: MySQL
- Backend Framework: Express.js
- ORM: Sequelize
- Frontend: HTML, CSS, JavaScript, Bootstrap
- Tools: VS Code, npm, Git, Postman (optional)

4. Data Diagram and Table

Data Flow Diagram (Level-0, text representation)

- User/NGO -> Care & Share System -> Database / File Storage -> Response to User/NGO
- Admin -> Care & Share System -> Database / File Storage -> Response to Admin

Core Table Summary

- users: id, email, password, firstName, lastName, isAdmin, accountType, ngoStatus, ngoDocuments
- products: id, name, category, type, price, productCondition, status, userId
- donate_items: id, itemName, itemType, quantity, itemCondition, status, userId
- donate_requests: id, donationId, receiverUserId, status, description
- exchange_requests: id, targetProductId, requesterId, exchangeItemName, status
- purchases: id, productId, buyerId, status, createdAt

5. ER Diagram (text representation)

- USERS 1..* PRODUCTS
- USERS 1..* DONATE_ITEMS
- USERS 1..* DONATE_REQUESTS
- USERS 1..* EXCHANGE_REQUESTS
- USERS 1..* PURCHASES
- DONATE_ITEMS 1..* DONATE_REQUESTS
- PRODUCTS 1..* EXCHANGE_REQUESTS
- PRODUCTS 1..* PURCHASES

Appendix A: Data Flow Diagram (Mermaid)

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flowchart TD
    U[User/NGO] -->|Register/Login, Add Items, Requests| S[Care & Share System]
    A[Admin] -->|Approve/Reject, Manage Users/NGOs| S
    S --> DB[(MySQL Database)]
    S --> FS[(File Storage: uploads)]
    DB --> S
    FS --> S
    S -->|Responses, Listings, Status| U
    S -->|Dashboards, Queues, Reports| A

```

Appendix B: ER Diagram (Mermaid)

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erDiagram
    USERS ||--o{ PRODUCTS : lists
    USERS ||--o{ DONATE_ITEMS : donates
    USERS ||--o{ DONATE_REQUESTS : receives
    USERS ||--o{ EXCHANGE_REQUESTS : requests
    USERS ||--o{ PURCHASES : buys
    DONATE_ITEMS ||--o{ DONATE_REQUESTS : requested_by
    PRODUCTS ||--o{ EXCHANGE_REQUESTS : target_product
    PRODUCTS ||--o{ PURCHASES : purchased_product

```